

Project Initialization and Planning Phase

Date	27 june 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

The **Project Initialization and Planning Phase** is the foundation of the Smart Agricultural Production Optimization Engine (OptiCrop). In this phase, the team identifies the major problems faced by farmers in crop selection and agricultural planning.

Farmers often struggle to choose suitable crops due to changing soil and environmental conditions such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall.

This project aims to solve these challenges by developing an intelligent system that analyzes agricultural data and provides crop recommendations. The planning phase helps define customer problems, project goals, scope, and expected outcomes. It ensures that the team clearly understands the user requirements and builds an efficient solution for improving crop yield and farming productivity.

Objectives:

- Understand farmer problems in crop selection
- Analyze agricultural data requirements
- Plan project scope and features
- Develop a smart decision-making system for farmers

Expected Outcome:

The project will provide accurate crop recommendations and improve agricultural productivity using data-driven insights.

Example:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A farmer	Select the best crop for farming	I don't know which crop is suitable for my soil and climate	Soil nutrients and environmental conditions vary frequently.	Confused about crop selection and productivity
PS-2	An agricultural researcher	Analyze crop-environment relationships	Large agricultural data is difficult to process manually	Many factors affect crop production	Challenged in making accurate decisions